Team 105

Mapping Climate Extremes and Economic Vulnerabilities across the United States

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Motivation

WHAT

Extreme weather events are becoming more frequent and costly, but existing tools show them separately, making it hard to understand overall county-level risk and recovery.

WHY

Communities need **clear, integrated information** to prepare for disasters and allocate resources fairly.

Data Sources

Downloaded from **public sources**: NOAA (NCEI Storm Events), FEMA Disaster Aid, Zillow housing metrics, BLS unemployment, Census population data using automated Python scripts and web scraping for bulk retrieval and cleaning.

Details:

- 1.3M NOAA storm event records (3.4 GB)
- 1M monthly Zillow market estimates (102 Mb)
- 24k FEMA distributions (2 Mb)
- 81k annual unemployment rates (3 Mb)
- 53k census populations (1.6 Mb)

Evaluation

1. We evaluated our risk ratings' s correlation with Zillow's "Market Heat Index"
   - Our index explains 13% of the variance in the Heat Index in an OLS model, with a statistically significant coefficient.
2. We also performed a case study using Hurricane Katrina, and compared it to an existing study on that storm's lasting effects
   - Our index highlights gulf coast states, particularly in Louisiana, immediately following the storm, agreeing with prior work
3. We outperform prior indices (CERI, BRIC) by covering multiple hazards, offering spatial-temporal resolution, and providing an interactive and intuitive tool.
4. We built upon existing resilience metrics with an exploratory decision-support tool.

Our Approach

WHAT

A composite risk index for each county each year that describes susceptibility to extreme weather, displayed on an interactive D3 choropleth.

HOW

1. Merge county level data by FIPS code and month-year
2. Normalize damages and aid per capita, adjust for inflation
3. Rank negative outcomes and average rankings to produce index

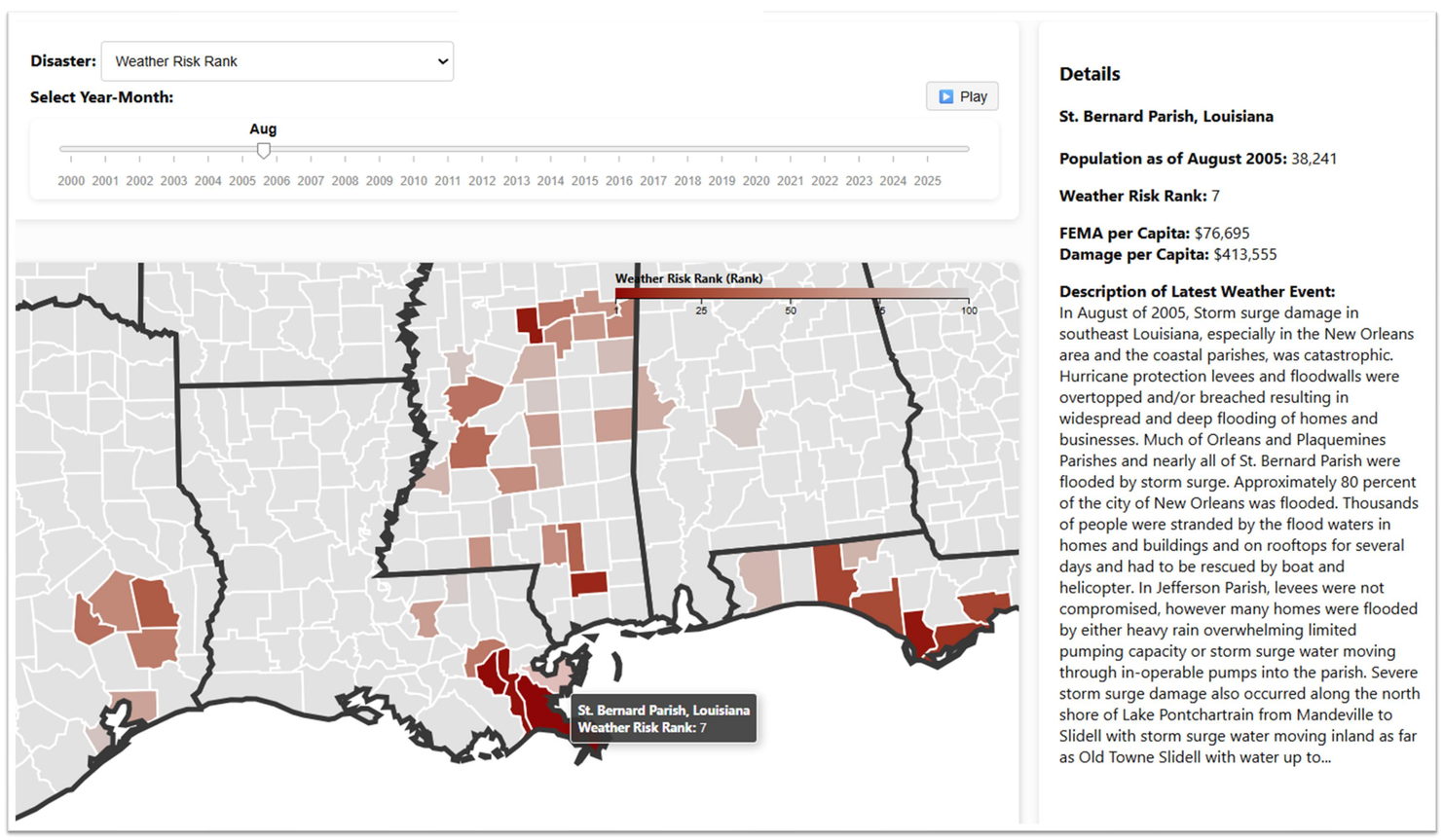
WHY

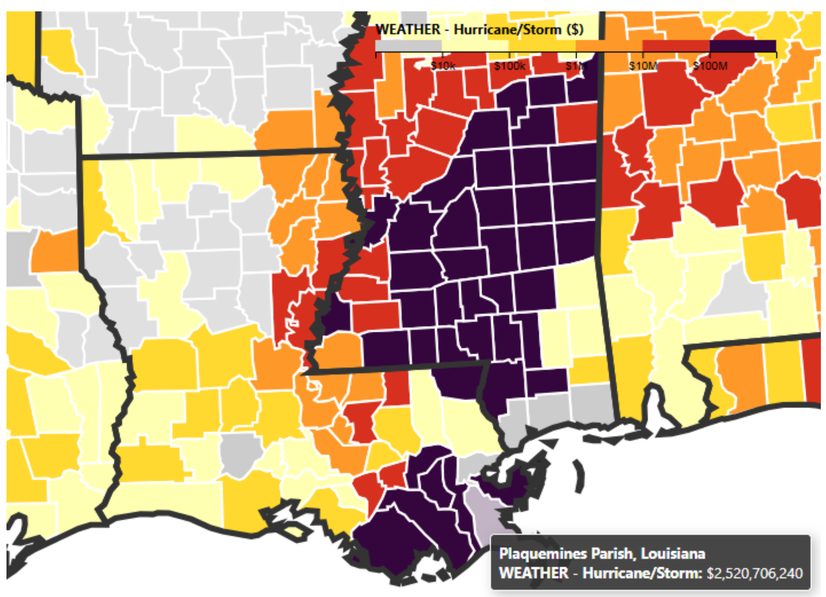
Climate risk is growing in many areas and needs to be considered wholistically. Fair normalized comparisons across counties are needed to highlight relative risk and display evolving dynamics of our climate and economy. Give users intuitive and actionable metrics in an interactive format is much more digestible than research papers alone.

WHAT IS NEW

Capturing compound climate and economic vulnerability is not easily accessible today. A longitudinal study over the past 25 years captures changing conditions across the country. And per-capita and inflation adjusted metrics remove noise and help focus attention on weather-specific impacts.

$$R_{c,t} = \text{rank}\left(\frac{1}{4}\sum_{i=1}^{4} r_i(c,t)\right),$$





Our interactive visualization consists of three coordinated panes: **Controls**, **Map**, and **Details**. The *Controls* pane allows users to explore the data through dropdown menus selecting parameters such as deaths, injuries, financial damage, FEMA obligated funding, and specific weather events. It also includes a time slider (2020–2025) with a Play/Pause function to animate trends over time. The *Map* pane displays a zoomable U.S. county-level choropleth that dynamically updates its colors and legend based on the selected parameter and year. Finally, the *Details* pane presents county-specific information on mouse hover, enabling users to quickly interpret localized impacts.